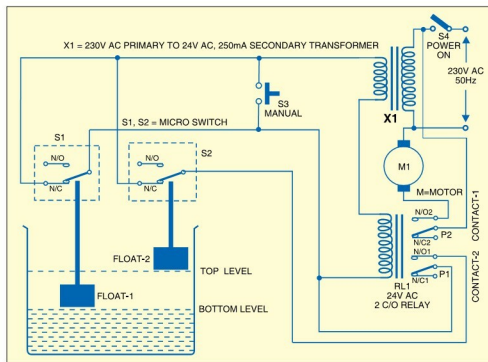




The relay should have two contacts. A step-down transformer having secondary voltage suited to the coil voltage of the relay is used. As the circuit works off AC, no rectification is necessary. Micro-switches S1 and S2 fixed on top of the water tank are operated by separate floats: one for sensing the bot-

tom level and the other for top level. A three-core wire is used for connecting these switches to the relay.

Working of the circuit is simple. As water level in the tank goes down to reach the bottom, float-1 releases the moving contact of micro-switch S1, causing the circuit to complete and en-

ergise relay RL1. The pump gets power supply through relay contact-1 and starts running. At the same time, relay contact-2 also closes, which provides 24V AC through moving contact of switch S2 connected in the normally-closed (N/C) position.

As water level rises in the tank, the moving contact of micro-switch S1 is opened by float-1, but the pump continues to run as the circuit completes through switch S2 and relay contact-2.

When the tank is full, the contact of switch S2 is opened by float-2 and the relay de-energises to stop the pump. The pump restarts only when water goes down to the bottom level and the moving contact of switch S1 is released by float-1. This cycle repeats and water level in the tank is controlled automatically.

If the mains power fails when the pump is running, the pump stops. It has to be restarted manually when power resumes. For this purpose, a push-to-on switch is provided. Align float-1 and float-2 such that these touch the moving contacts of switches S1 and S2 when water touches the respective level. ●

OVER-HEATING INDICATOR FOR WATER PIPE

■ M.K. CHANDRA MOULEESWARAN,
S. SANTHOSH KUMAR

The hot water pipe from the water geyser of your bathroom may burst if it gets overheated and is left unattended. This circuit monitors the temperature of the water pipe. If the temperature of the pipe goes above certain limit, it flashes an LED.

The circuit uses a small sensor with inbuilt thermostat. It is actually an automatically resettable thermal switch that operates whenever its temperature exceeds the set limit. It is available in two versions, viz, normally open (N/O) and normally closed (N/C).

The N/O version closes the switch upon the set limit, while the N/C ver-

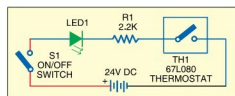


Fig. 1: Circuit for showing over-heating status of the pipe through LED

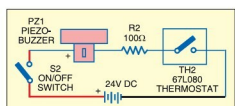


Fig. 2: Circuit for aural indication of over-heated pipe



sion opens the switch. The thermostat automatically resets as soon as the temperature returns to the initial state. It can be directly mounted on the part to be monitored.

The sensor is available in various temperature settings. The temperature limit is specific from 70°C up to 110°C for each model number. For instance, the type 67L080 will change state at around 80°C±5°C. You can select the type suitable for your requirement. Use a three-core wire for connecting switches to the relay.

The circuit uses very few components. Also, it does not require a separate power supply and can be connected to the power supply of the